

Ravi Singh

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SUMMARY

Senior Software Engineer with **6+ years of experience building highly scalable, test-driven backend systems** and production-grade **GenAI platforms**. Proven expertise in **event-driven architectures**, **message queues**, and **distributed services**, combined with hands-on experience building **AI agents**, **RAG pipelines**, and **vector-search systems** using LangChain, LangGraph, LlamaIndex, and Qdrant. Focused on **reliability**, **maintainability**, and **long-term system health**.

EXPERIENCE

EPAM Systems Inc.

Senior Software Engineer

Oct 2023 – Present

Noida, India

Scalable Backend & Platform

- Designed and built **highly scalable backend services** to support AI-driven customs classification workflows.
- Implemented **event-driven architectures** using **RabbitMQ** to decouple long-running classification, reporting, and reprocessing jobs.
- Defined **clear service boundaries and API contracts**, improving maintainability and independent service evolution.
- Applied **TDD for core business logic**, writing unit and integration tests to ensure correctness and long-term reliability.

GenAI & Agent Systems

- Built **GenAI-powered agents** using **LangChain and LangGraph** to orchestrate multi-step workflows such as classification, validation, and reporting.
- Implemented **RAG pipelines** using **LlamaIndex** with **Qdrant** as a vector database for semantic search over internal knowledge and historical classifications.
- Designed prompt strategies and agent routing logic to safely execute backend actions from natural language inputs.
- Integrated **MCP (Model Context Protocol)** to standardize context sharing across tools and agents.

Frontend

- Built conversational UIs that interact with backend agents via Streaming APIs.
- Improved frontend performance with caching, lazy loading, and optimized data-fetching, reducing backend load and improving UX.
- Improved frontend performance through **caching strategies**, **lazy loading**, and **code splitting**, resulting in **~10% faster load times** and **20% smaller bundles**.

Tesit Technologies Inc.

Software Engineer

Feb 2021 – Oct 2023

Remote

- Led development of a **high-performance PWA** using **Next.js**, achieving **99%** Lighthouse performance and **100%** SEO.
- Designed and built **scalable video-processing microservices** using **Node.js** and **FFmpeg**, supporting **~10k/day users**.
- Containerized services with **Docker** and deployed **Kubernetes auto-scaling** to handle traffic spikes.
- Built AI-driven video auto-tagging pipelines using **AWS Kinesis**, improving content discovery.
- Delivered core user experiences including **personalized feeds**, **in-app video editing**, and **real-time messaging**.
- Improved developer velocity by creating **custom GitHub Actions** for CI/CD.

Envirya Technologies Pvt. Ltd.

Software Engineer

Aug 2019 – Feb 2021

Gurgaon, India

- Built a **custom, reusable form builder** in **React Native** used across multiple projects.
- Developed MSME-focused mobile and backend solutions using **React Native**, **Node.js**, and **MongoDB**.
- Automated operational workflows using background services and **Cron jobs**.
- Developed RESTful backend services using **Node.js** to support web and mobile applications.

SKILLS

Backend: Node.js, Express, Python, FastAPI, REST, GraphQL, gRPC, WebSockets

Messaging & Async Processing: RabbitMQ, Background Workers, Retry & Idempotency Patterns

Frontend: React, Next.js, TypeScript, Tailwind, Material UI, Shadcn

Testing: TDD, Jest, React Testing Library, Cypress, Playwright, API Testing, Unit & Integration Testing

AI & Automation: LangGraph, LangChain, MCP, LlamaIndex, Prompt Engineering, AWS Bedrock Flows

Cloud & DevOps: GCP, AWS (Lambda, S3, Bedrock), Docker, Kubernetes, Jenkins, CI/CD

Databases: MongoDB, Qdrant, Semantic Search, Embeddings, PostgreSQL, MySQL, Big Query

EDUCATION

Bachelor Of Engineering – Computer Science and Engineering

2015 – 2019

Dr. B. R. Ambedkar University, Agra